XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

DESIGN AND IMPLEMENTATION OF A SECURE SELF-HOSTED CLOUD STORAGE SYSTEM

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OVERVIEW

01 Introduction

02 Problem Statement

03 Aim & Objectives

04 Literature Review

05 Project Contribution

06 Methodology

07 System Features

08 Project Significance

09 Expected Outcomes

10 Testing

11 Gantt Chart

12 Conclusion

01

INTRODUCTION

Cloud services offer convenience but often compromise privacy and control. This project presents a user-friendly, secure self-hosted cloud solution that enables small teams and individuals to manage their own data, ensuring autonomy without sacrificing usability.

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PROBLEM

01 Centralized cloud services collect, archive, and monetize user data (Ali et al., 2018; Ullah et al., 2020)

02 Users have limited control over privacy settings and data policies (Zeng et al., 2022; Obar & Oeldorf-Hirsch, 2020)

03 Vendor lock-in limits user freedom and portability (Martins et al., 2016; CVL, 2024)

04 Open-source alternatives exist but are too complex for average users (Wu & Chen, 2022; El-Gazzar et al., 2019)

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AIM

Develop an accessible, privacy-focused self-hosted cloud storage system based on an open-source platform (e.g., Seafile), with enhanced usability and deployment features.

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OBJECTIVES

01 Examine the architecture and operation of the selected open-source cloud storage system and expand it by adding meaningful custom features (e.g. permission presets, optional encrypted uploads, deployment helper scripts)

02 Develop a comparative evaluation framework to benchmark the prototype against popular commercial (Google Drive, Dropbox) and open-source (Nextcloud, Seafile) systems, focussing on usability, privacy control, and ease of self-hosting.

03 Produce comprehensive system documentation and modular source code to facilitate reuse and learning by students, small teams, or hobbyists wanting to build or modify their own cloud storage system.

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LITERATURE REVIEW

- Centralized platforms suffer from privacy breaches and vendor lock-in (Ali et al., 2018; CVL, 2024)
- Most users cannot understand complex privacy agreements (Obar & Oeldorf-Hirsch, 2020)
- Open-source tools like Nextcloud and Seafile offer control but require high technical skills (Wu & Chen, 2022; El-Gazzar et al., 2019)
- There is no perfect balance between usability, privacy, and extensibility in existing platforms

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PROJECT CONTRIBUTION

01 Develop a hybrid cloud solution that balances ease-of-use and privacy.

02 Lower the technical barrier for self-hosting through automation and presets.

03 Provide a deployable system and comparative evaluation for learning.

04 Contribute to ethical and inclusive digital infrastructure.

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METHODOLOGY

Methodology Research Design:

- Quantitative, based on Design Science Research

Data Collection:

- Technical tests: upload/download speed, sync time, resource usage
- Questionnaire-based usability testing (Likert scale)

Analysis:

- Descriptive statistics, comparison charts

Phase 01

Requirement Analysis & Platform Selection

Phase 02

System Development & Enhancement

Phase 03

Evaluation & Benchmarking

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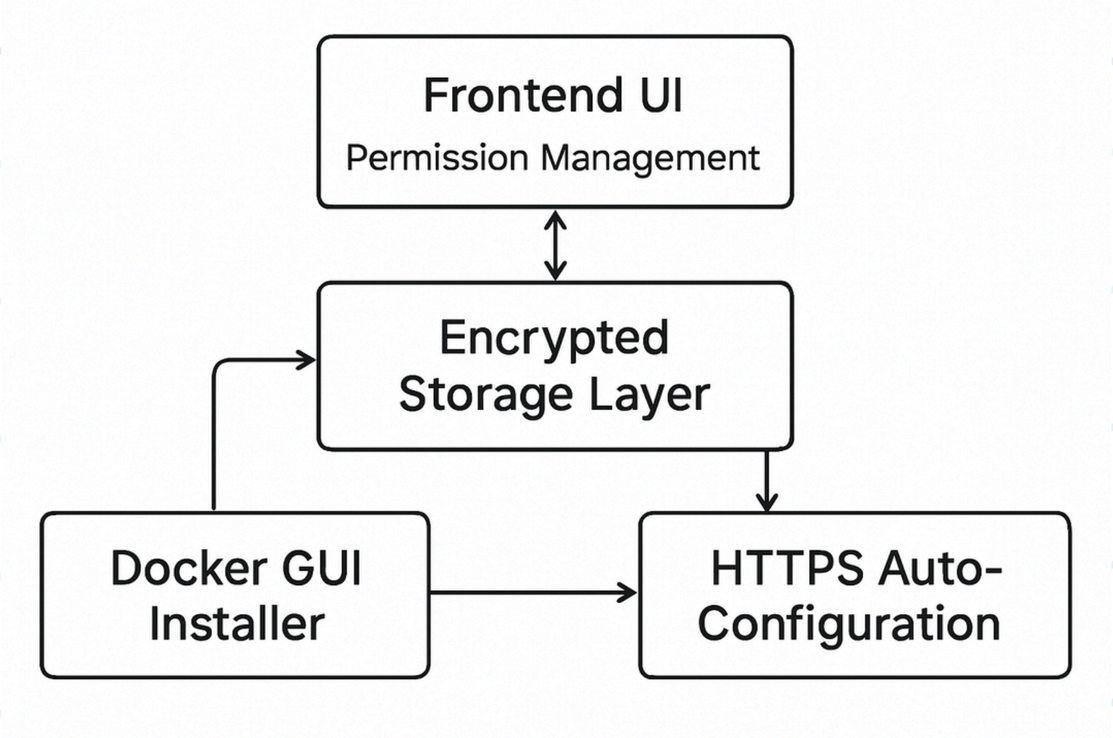
SYSTEM FEATURES (ARCHITECTURE OVERVIEW)

- Front-end UI with permission management
- Encrypted storage layer
- Docker GUI installer simplifies setup
- HTTPS auto-configuration



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PROJECT SIGNIFICANCE

- Enables small organizations and individuals to reclaim data control
- Reduces dependence on commercial cloud providers
- Supports digital literacy and self-hosting knowledge
- Serves as a replicable educational model for privacy-conscious systems

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EXPECTED OUTCOMES

- A deployable Seafile-based cloud storage system
- Automated setup scripts and user-friendly configuration
- Optional privacy features like encryption and permission presets
- Documentation for installation and adaptation

TESTING

Technical Testing:

- Upload/download speed (measured in Mbps)
- Sync performance across multiple users and file types
- CPU/RAM resource benchmarks

User Testing:

- 8–10 participants complete usability tasks
- Questionnaire on installation, interface, satisfaction, perceived privacy
- Analysis of scores to compare against commercial services

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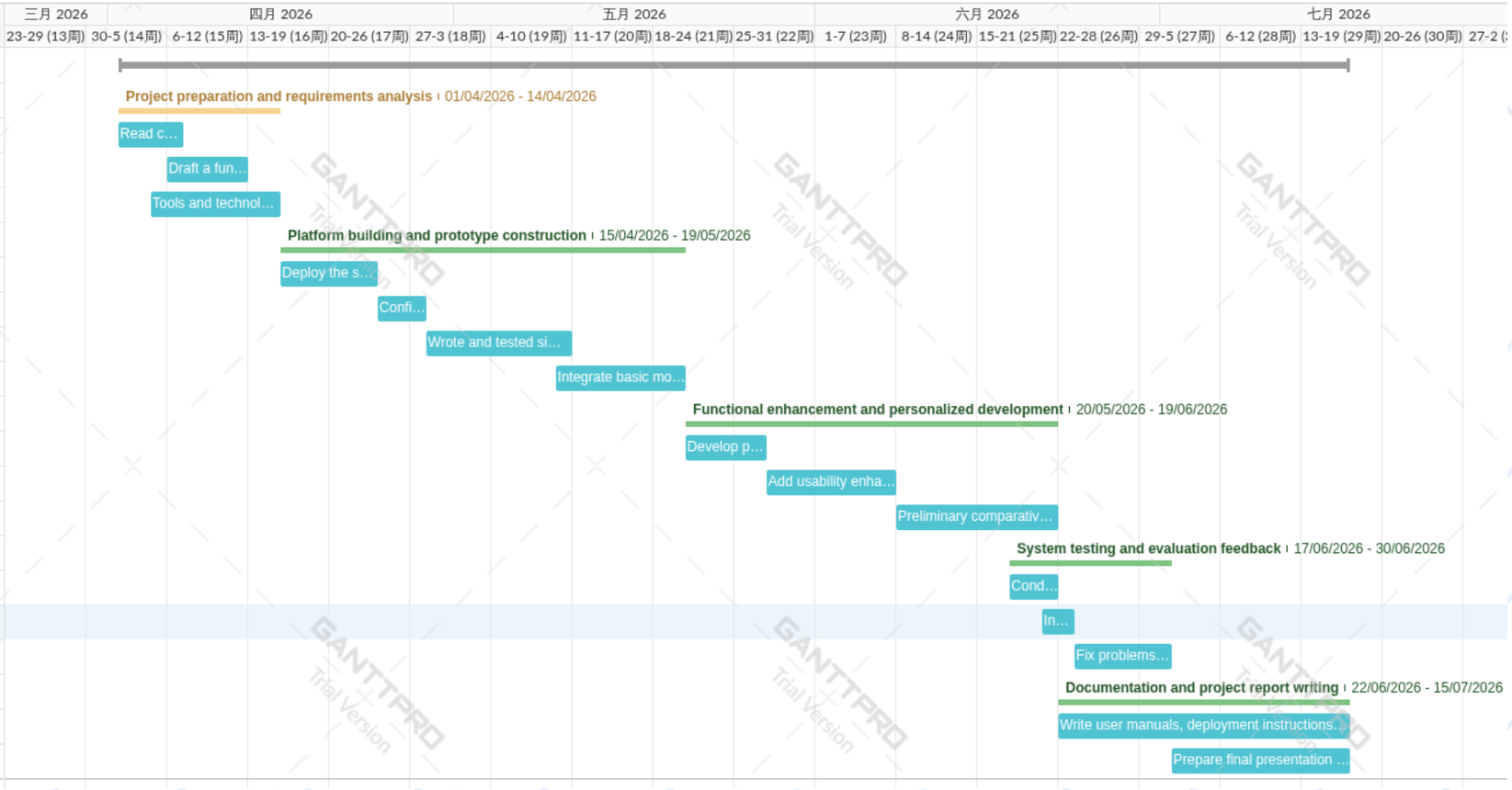
GANTT CHART

	任务名称
1	**Project preparation and requirements analysis**
1.1	Read core literature and research open source platforms (Nextcloud, Seafile, Cloudreve, etc.)
1.2	Draft a functional requirements list and preliminary system architecture design
1.3	Tools and technology stack determination
2	**Platform building and prototype construction**
2.1	Deploy the selected platform locally or on a VPS (preliminary Nextcloud + Docker solution)
2.2	Configure the basic environment (database, SSL, Web service)
2.3	Wrote and tested simplified deployment scripts (automatic SSL, database initialization)
2.4	Integrate basic modules such as login, file management, and permission settings
3	**Functional enhancement and personalized development**
3.1	Develop privacy-enhancing features (e.g., optional end-to-end encryption, permission templates)
3.2	Add usability enhancements (such as installation guide interface, interface simplification)
3.3	Preliminary comparative analysis with commercial platforms (deployment, functionality, UI, privacy)
4	**System testing and evaluation feedback**
4.1	Conduct comprehensive testing (functional testing, safety testing, performance testing)
4.2	Invite a small group of users to try it out and collect feedback (such as instructors and classmates)
4.3	Fix problems and perform iterative optimization
5	**Documentation and project report writing**
5.1	Write user manuals, deployment instructions, and system architecture documents
5.2	Prepare final presentation materials (PPT)



CONCLUSION

This project delivers a practical, self-hosted cloud storage system that is secure, user-friendly, and accessible. By reducing technical friction and improving privacy controls, it empowers users to manage their own data independently and confidently.

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THANK YOU